

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION
DURATION: 3 HOURS

SUMMER SEMESTER, 2021-2022
FULL MARKS: 150

CSE 4203: Discrete Mathematics

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 6 (six) questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

- ✓ 1. Consider Equation 1 and answer the following:

$$\frac{-b + \sqrt{b^2 - 4ac}}{2a} \quad (1)$$

- | | |
|--|--------------------------|
| a) Generate the corresponding rooted tree. | 5
(CO1)
(PO1) |
| b) Generate the corresponding prefix and postfix notations. (Avoid using decimal fractions) | 5 × 2
(CO2)
(PO1) |
| c) Evaluate the postfix and the prefix notations. | 5 × 2
(CO2)
(PO1) |
| ✓ 2. a) How does Huffman coding resolve the issues associated with simple prefix codes? Explain with an example. | 5
(CO2)
(PO1) |
| b) Use Huffman Coding to generate the prefix codes of the characters in the string, "make each day your masterpiece". Calculate the average number of bits required to represent each character using the generated prefix codes.
Ignore spaces and consider arranging characters with the same probability in lexicographic order. | 15 + 5
(CO2)
(PO1) |
| ✓ 3. a) Use the principle of Mathematical Induction to verify that, $\forall n \in \mathbb{Z}^+, 6^n - 1$ is divisible by 5. | 7
(CO1)
(PO1) |
| b) Using the binary addition algorithm, show step-by-step process of adding the two binary numbers, $(101010)_2$ and $(111111)_2$. | 8
(CO1)
(PO1) |
| c) Consider that the two functions, $f(x)$ and $g(x)$, defined by $f(x) = \frac{4x-3}{x-2}$ and $g(x) = \frac{2x+4}{x-4}$, respectively that follow the relation, $\mathbb{Z}^+ \rightarrow \mathbb{Z}^+$. Find $f^{-1}(x)$ and $f \circ g(x)$. | 5 + 5
(CO1)
(PO1) |

4. The fictional country of Wakanda has both Railways and Waterways for transporting food crops via land and sea, respectively. There are a total of 16 ports, 8 of which are located in the coastal areas and the rest are located inland. The ports in the respective geographies are graphically connected as shown in Figure 1, where each of the land ports and seaports are denoted by R_i and W_j , respectively. Based on this information, answer the following -

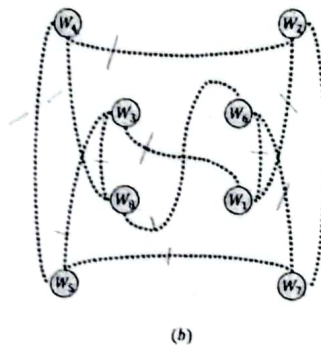
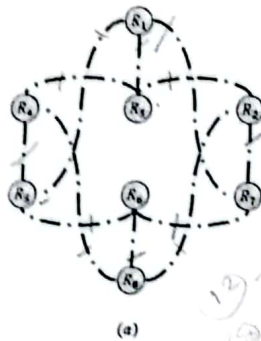


Figure 1: Graphs representing the connection between the eight (a) land ports and (b) seaports of Wakanda for Question 4.

- (a) Analyze the graphs in Figure 1 and determine whether the two graphs are isomorphic. 12
(CO3)
(PO1)
- (b) Apply Dirac's and Ore's theorem to determine the presence of Hamiltonian Circuit(s) in both of the graphs in Figure 1. 13
(CO3)
(PO1)
5. (a) Use rules of inference to show that the premises, "A student in this class has not read the book" and "Everyone in this class passed the first exam" imply the conclusion, "Someone who passed the first exam has not read the book". 13
(CO2)
(PO1)
- b) Give a big-O estimate for $f(n) = (n + 1) \log(n^2 + 1) + 3n^2$, where $n \in \mathbb{Z}^+$. 12
(CO2)
(PO1)
6. a) Suppose a person deposits BDT 10,000 in a savings account at a bank yielding 11% per year with interest compounded annually. How much will be in the account after 30 years? 9
(CO3)
(PO1)
- b) Consider the message, "LIFE IS BEAUTIFUL WITH DISCRETE MATHEMATICS." Now encrypt it (including spaces) by applying transposition cipher with $\sigma = \{1, 2, 3, 4\}$ and $\sigma(1) = 2, \sigma(2) = 4, \sigma(3) = 1$, and $\sigma(4) = 3$. 8
(CO1)
(PO1)
- (c) Give a recursive algorithm for computing the GCD (Greatest Common Divisor) of two non-negative integers a and b with $a < b$, where $a = 19$ and $b = 57$. Show step-by-step execution of your algorithm. 8
(CO3)
(PO1)